Ecological Modelling 486 (2023) 110509

Contents lists available at ScienceDirect

Ecological Modelling

journal homepage: www.elsevier.com/locate/ecolmodel

Development of a pelagic biogeochemical model with enhanced computational performance by optimizing ecological complexity and spatial resolution

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ARTICLE INFO

Keywords:
Eutrophication
Ecological modelling
Nutrient cycling
Dinitrogen fixation
Curonian Lagoon

ABSTRACT

Introducing ESTAS-AQUABC, a new software tool designed for modelling nutrient cycles and eutrophication in highly-eutrophic estuaries. Our focus on replicating intense cyanobacteria blooms and N2-fixation sets us apart from other estuary models. We account for N2-fixation by cyanobacteria, the inorganic carbon cycle, non-conservative alkalinity, and pH calculations through CO2SYS integration. AQUABC also includes ammonia-stripping at high pH in its kinetic sub-model. ESTAS-AQUABC uses box modelling approach where the fluxes between boxes are calculated by a hydrodynamical model, in our case the SHYFEM model. This approach is robust, especially when field data for calibration is scarce, saving computational time. While our model accurately reproduced pelagic N2-fixation rates, inorganic nutrient dynamics were less precise, mainly due to the absence of bentho-pelagic exchange processes. ESTAS-AQUABC provides a useful tool to investigate nutrient retention from land to open sea, which could be important to mitigate eutrophication in estuarine systems and the adjacent coastal areas.

1. Introduction

Coastal eutrophication is a global problem induced by intensive nutrient loads from catchment to downstream aquatic systems (Boesch et al., 2006; Carpenter et al., 2007; Garmendia et al., 2012). Consequently, it leads to the eutrophication manifested by excessive primary production, toxic algal blooms, oxygen depletion, sulphide accumulation and loss of biodiversity (Bonsdorff et al., 1997; Conley et al., 2011). Since estuaries are complex and heterogeneous systems hosting multiple processes, the application of models appears to be most useful to conceptualize, integrate and generalize the eutrophication causes and consequences.

Incorporation of nutrient cycles into water quality models necessitated the introduction of new chemical/biochemical processes and state variables, such as ammonia, nitrates, phosphates, organic nitrogen, phosphorus, and phytoplankton biomass. Developments in computer technology enabled scientists and engineers to design and develop these models, which included the models used by Di Toro et al. (1971), Thomann et al. (1975) and Di Toro and Connolly (1980). These studies were the first ones that utilized numerical models for analysis of eutrophication process. The models developed in these studies were the predecessors of most of the recently used nutrient dynamic modelling tools.

A spatially highly resolved multidimensional model is needed to analyse the eutrophication process in large water bodies that are generally horizontally and/or vertically not well mixed so that environmental gradients are expected. Developing (spatially) high-resolution multidimensional model is a challenge where several issues have to be addressed. Many studies conducted in the last two decades utilized dynamic multidimensional eutrophication models with a high spatial resolution (e.g. Arhonditsis et al., 2000; Drago et al., 2001; Rukhovets et al., 2003; Chau, 2004; Hartnett and Nash, 2004; Jakobsen et al., 2007; Menesguen et al., 2007; Vanderborght et al., 2007; Chao et al., 2010; Timmermann et al., 2010; He et al., 2011; Fernandez et al.,

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https://doi.org/10.1016/j.ecolmodel.2023.110509
Received 24 January 2023; Received in revised form 7 September 2023; Accepted 20 September 2023
Available online 2 October 2023
0304-3800/

2012; Zouiten et al., 2013; Zhu et al., 2016; Baustian et al., 2018). First of all, all the water fluxes entering and leaving the computational elements of the eutrophication model must be generated. There are two basic approaches to implement this task; the coupled and the decoupled hydrodynamic forcing. The first approach is the "direct" or "online" coupling where the hydrodynamic and eutrophication model codes are merged into one single software. Hydrodynamic transport is calculated, scalars are transported and biogeochemical equations are solved simultaneously. In this approach, feedback effects between biogeochemistry and hydrodynamics (such as effects of change of light transmission or viscosity because of the algal blooms on hydrodynamics) can be simulated as well. Studies by Butenschon et al. (2012) and Umgiesser et al. (2003) are examples for the first approach. The second approach is the "offline coupling", where the hydrodynamic simulation is conducted first and once validated, the results from the hydrodynamic model are read into the ecological model. Models such as WASP (Wool et al., 2020) or AQUATOX (Park and Clough, 2018) follow this approach. The advantages and limitations of both approaches are discussed by Kim and Khangaonkar (2012). Basically, the coupled hydrodynamic forcing requires more computational power, whereas the decoupled hydrodynamic forcing requires more storage space. In most cases, the first approach is favored while also a considerable number of studies are conducted using the second approach (Drago et al., 2001; Rukhovets et al., 2003; Lee et al., 2005; Dortch et al., 2008; Cerco et al., 2010a; Kim and Khangaonkar, 2012). Clearly, the second approach facilitates the calibration and validation of the ecological model, because many more simulations can be carried out much faster.

The complexity of the ecological sub-models is also important as it decides how realistically the real-world processes are resolved. There are different complexity level models used from a few to dozens of state variables that include multiple phyto- and zooplankton groups, sediment diagenesis, and submerged aquatic vegetation (Hartnett and Nash, 2004; Jakobsen et al., 2007; Vanderborght et al., 2007; Cerco et al., 2010a; Fernandez et al., 2012). Limited number of state variables, such as for example a single zooplankton group, restricts simulating only a single food-chain (Lopes et al., 2010; Timmermann et al., 2010). Including more phytoplankton or zooplankton groups a model would address the different affinity for nutrients for phytoplankton groups and different preferences for phytoplankton in zooplankton groups, facilitating competition and seasonal succession.

In this study, we have developed a pelagic nutrients-phytoplankton-zooplankton-detritus (NPZD) box-model ESTAS-AQUABC and tested it on the largest European lagoon, the Curonian Lagoon. In comparison to our previous modelling study for this lagoon (Zemlys et al., 2008) when developing this model, we aimed for a finer resolution of the nutrient cycling, with a particular focus on atmospheric dinitrogen (N_2) fixation by diazotrophic cyanobacteria. The model resolves the inorganic carbon cycle and alkalinity as a state variable and calculated pH using CO2SYS (Lewis and Wallace, 1998). These processes are incorporated in many leading model frameworks with a focus on ocean acidification. Since application of AQUABC (Aquatic Biogeochemical Cycles) is focused in estuarine areas incorporation of inorganic carbon cycle, alkalinity and pH will provide a realistic simulation of the freshwater dominated watershed to ocean continuum through the estuaries and coastal lagoons.

The underlying hydrodynamic model used in this study consists of the SHYFEM hydrodynamic modelling framework (http://www.ismar.cnr.it/shyfem) which solves the hydrodynamic equations on a finite element model domain. Even though internally coupling of the kinetic model (AQUABC) with SHYFEM will be available for future studies (to the same extent as other biogeochemical models like EUTRO or BFM), here we have focused on a box model approach. The ecological model was linked "offline" with SHYFEM, meaning that all hydrodynamic variables were computed on boxes consisting of a variable number of finite elements. Hydrodynamic fluxes were computed between all these boxes, and variables and parameters were computed inside the boxes as average values of the boxes. Since SHYFEM runs on an unstructured grid, ESTAS-AQUABC model boxes can also be of irregular shape and therefore represent water bodies that have irregular shapes and/or complex coastlines with a reasonable number of computational elements. Therefore, this box model allows the construction of model domains with spatially varying spatial resolutions such as deep lakes and streams. This is particularly interesting as multiple biogeochemical processes in lagoons can largely affect the nutrient transport from the river basin to the Baltic Sea (Petkuviene et al., 2016; Vybernaite-Lubiene et al., 2017; Zilius et al., 2018). Elucidating the nutrient cycling in estuarine ecosystems is important for understanding nutrient attenuation, regulation of eutrophication, and for forecasting recovery from eutrophication.

2. Modelling framework

2.1. General description

The ESTAS-AQUABC, which is the computational engine of the modelling package developed in this study, is designed as a pelagic box model, where the boxes can be configured to generate spatially zero-, one-, two- and three-dimensional model domains representing small and well-mixed standing waterbodies, rivers, shallow and deep waterbodies. This ensures flexibility and consequently wider applicability to different waterbodies and scientific problems. One notable feature of the ESTAS-AQUABC approach is its reliance on lower spatial resolution. This means that the simulation does not require an extremely intricate representation of the physical environment, which can be particularly beneficial in situations where high-resolution data is scarce or unavailable. Instead, the model can work effectively with coarser spatial data, making it more robust when it is available only limited or unevenly distributed field measurement data. Moreover, the lower spatial resolution approach has a practical advantage in terms of computational efficiency. The calibration process, which is essential for ensuring that the model accurately reflects real-world conditions, is less computationally demanding when using lower-resolution data. This is a significant benefit, as it reduces the computational time required for model calibration and thus speeds up the overall modelling efforts.

The details of the box modelling approach are explained in many standard modelling texts such as Orlob (1983), Thomann and Mueller (1987) and Chapra (2008). In this study framework, the box model called ESTAS (Ecosystem and Transport Simulator) solves the advection-diffusion-reaction equation using the box modelling approach as the spatial discretization, whereas the eutrophication kinetics are handled by AQUABC pelagic ecology model. The modelling framework of ESTAS-AQUABC together with accompanying tools is summarized in Fig. 1.

The full list of state variables of AQUABC are given in Table 1.

The box model includes the following components, all of which have a tabular input data of time and space:

- *The advective transport field* accounts for the flows between the boxes.
- *The diffusive transport field* accounts for the turbulent transport between the boxes.
- *Settling* accounts for the gain or loss of material between the boxes and losses from the pelagic system.
- *Boundaries* – user can prescribe the water exchange between the model domain and its environment; and boundary concentrations on each open boundary.
- *Mass withdrawal time* series from the system represent zero order negative loads given in mass/time dimensions to account for any prescribed withdrawal of materials (as state variables) that do not have an attached water flux (a prescribed effect of filter-feeders such as mussels, prescribed fluxes because of diffusion into the sediments, etc.)
- *Kinetics* corresponds to biogeochemical cycles.

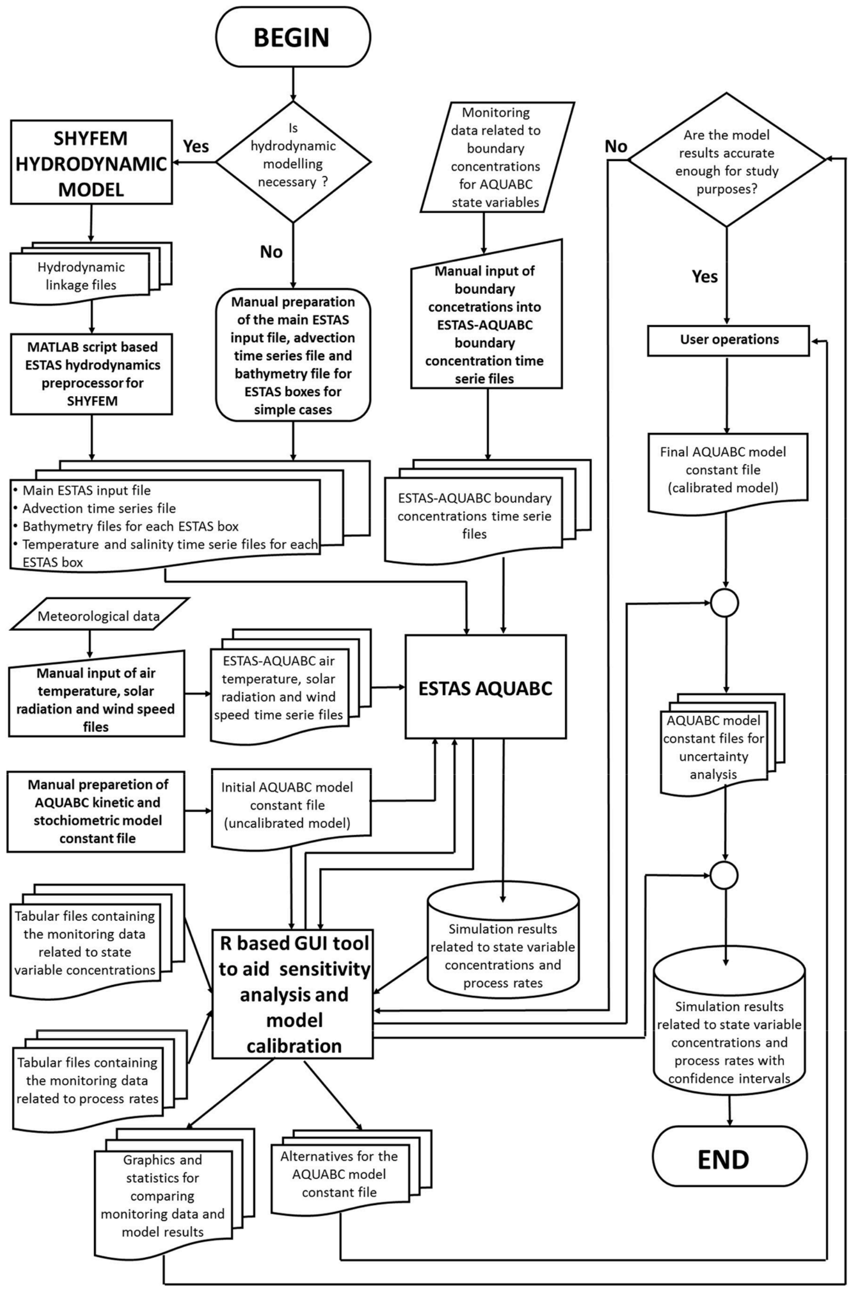

Fig. 1. The modelling framework.

- *Resuspension* can be either directly prescribed as time series of resuspension velocities or resuspended concentrations corresponding to state variables (for example intensive monitoring based time series if available or from a linked external model which produces resuspension time series) or the simple approach utilizing the internal facilities in the model, which compare externally read shear stress time series and compare them with prescribed critical shear stresses to switch on and off the settling process when resuspension of the "fluffy layer" occurs (i.e., when the calculated shear stress exceeds the critical shear stress).

Considering the components, the model assembles the set of equations (advection-diffusion-reaction equation) given in the Eqs (1):

$$\frac{\partial C_{n,i}}{dt} = \underbrace{\sum_{j=1*}^{\substack{\textit{Number of}\\ \textit{neighbouring}\\ \textit{boxes to box } i}} \frac{Q_{j,i}}{V_i}\cdot C_{n,j}}_{\substack{\textit{Inflow from neighbouring}\\ \textit{boxes}}} - \underbrace{\sum_{j=1}^{\substack{\textit{Number of}\\ \textit{neighbouring}\\ \textit{boxes to box } i}} \frac{Q_{j,i}}{V_i}\cdot C_{n,i}}_{\substack{\textit{Outflow to neighbouring}\\ \textit{boxes}}} + \underbrace{\sum_{j=1}^{\substack{\textit{Number of}\\ \textit{neighbouring}\\ \textit{boxes to box } i}} \frac{A_{i,j}\cdot D_{i,j}}{\ell_{i,j}\cdot V_i}\cdot\left(C_{n,j} - C_{n,i}\right)}_{\substack{\textit{Diffusion between box } i \textit{ and the}\\ \textit{neighbouring boxes}}} +$$
$$\underbrace{\frac{v_{set,overlaying,i}\cdot A_{i,overlaying}}{V_i}\cdot C_{n,overlaying\ box}}_{\substack{\textit{Settling from the overlaying box into the}\\ \textit{box } i}} - \underbrace{\frac{v_{set,i}\cdot A_{i,underlaying\ box}}{V_i}\cdot C_{n,i}}_{\substack{\textit{Settling from the box } i \textit{ into the}\\ \textit{underlaying box or out of the system}}} + \tag{1}$$
$$\underbrace{\sum_{m=1}^{\substack{\textit{number of}\\ \textit{external}\\ \textit{sources and}\\ \textit{sinks}}} S_{m,n,i}}_{\textit{Boundary forcing}} + \underbrace{\sum_{k=1}^{\substack{\textit{number of}\\ \textit{kinetic processes}\\ \textit{for the state}\\ \textit{variable}}} R_{k,n,i}}_{\textit{Kinetics, AQUABC in this case}}$$

where index i corresponds to the actual box, index j corresponds to any other box neighbouring box i, $C_{n,i}$ and $C_{n,j}$ are the concentrations for state variable n for box i and j respectively, $Q_{j,i}$ is the flow rate from box j to box i [$L^3 \cdot T^{-1}$], $Q_{i,j}$ is the flow rate from box i to box j [$L^3 \cdot T^{-1}$], $D_{i,j}$ is the diffusion coefficient between boxes i and j [$L^2 \cdot T^{-1}$], $\ell_{i,j}$ is the mixing length between boxes i and j [L], $A_{i,j}$ is the interface area between boxes i and j [L^2], V_i is the volume of box i [L^2], and $v_{set,i}$ is the settling velocity at box i, $S_{m,n,i}$ is the external source indexed as m related to any state variable indexed as n for box i accounting for boundary exchange, mass loads, mass withdrawals and resuspension [$M \cdot L^{-3} \cdot T^{-1}$] and the $R_{k,n,i}$ is kinetic reaction rate k for state variable n for box i [$M \cdot L^{-3} \cdot T^{-1}$]. Since the numerical solving of transport equations and process rate equations require higher computational performance Fortran 2003 was used to develop the computational engine code in order to produce a model that would run acceptably fast even for fairly complex model domains. There are several free optimizing Fortran 2003 compilers and the Fortran 2003 standard also supports vectorization. Calculation of AQUABC state variables was vectorised over the ESTAS boxes. ESTAS-AQUABC was successfully compiled and tested with Intel Fortran and gfortran compilers.

One important aspect to highlight is that the model is not conceived as a hydrodynamic model. Instead, the water fluxes within the model boxes need to be calculated or estimated externally using the SHYFEM hydrodynamic model, which considers the both mentioned forcing parameters. The output of the SHYFEM was formatted as time series compatible with ESTAS-AQUABC. The same approach applies to the handling of salinity and temperature. In the case of simple aquatic ecosystems, such as ponds or rivers, water fluxes, temperature, and salinity can be continuously monitored at appropriate intervals and provided to the model as input data. All external forcing is managed by the transport model and supplied to the pelagic ecology model. The required time resolution for external forcing depends on the nature of the case study. For example, an eutrophication study conducted in a small pond may only require monthly average water temperature and daily averaged meteorological information, while typical water quality modelling studies in inland waters may necessitate daily time series for all forcing factors. In the case of shallow coastal lagoons like the Curonian Lagoon, as demonstrated in this study, it is advisable to utilize hourly time resolution for all external forcing variables, including water fluxes, meteorological forcing, water temperature, and ice conditions. The modelling framework developed in this study is flexible enough to accommodate all these scenarios. It features a generalized object for time series that can be at any desired time resolution, and it also allows for the supply of different time series at different time resolutions.

Table 1
The list of state variables of the pelagic model.

State Variable No	State Variable	Notation	Unit
1	Ammonium nitrogen	[NH_4^+-N]	gN·m^{-3}
2	Nitrate nitrogen	[NO_3^--N]	gN·m^{-3}
3	Orthophosphate phosphorus	[PO_4^{3-}-P]	gP·m^{-3}
4	Dissolved silica	[$Si(OH)_4$-Si]	gSi·m^{-3}
5	Dissolved oxygen	[O_2]	gO_2·m^{-3}
6	Diatoms carbon	[Dia-C]	gC·m^{-3}
7	Non-fixing cyanobacteria carbon	[Cyn-C]$_{NOFIX}$	gC·m^{-3}
8	Fixing cyanobacteria carbon	[Cyn-C]$_{FIX}$	gC·m^{-3}
9	Other planktonic algae carbon	[OPA-C]	gC·m^{-3}
10	Zooplankton carbon	[Zoop-C]	gC·m^{-3}
11	Zooplankton nitrogen	[Zoop-N]	gN·m^{-3}
12	Zooplankton phosphorus	[Zoop-P]	gP·m^{-3}
13	Detrital particulate organic carbon	[POC]$_{DET}$	gC·m^{-3}
14	Detrital particulate organic nitrogen	[PON]$_{DET}$	gN·m^{-3}
15	Detrital particulate organic phosphorus	[POP]$_{DET}$	gP·m^{-3}
16	Particulate silica	[PSi]	gSi·m^{-3}
17	Dissolved organic carbon	[DOC]	gC·m^{-3}
18	Dissolved organic nitrogen	[DON]	gN·m^{-3}
19	Dissolved organic phosphorus	[DOP]	gP·m^{-3}
20	Dissolved inorganic carbon	[DIC]	gC·m^{-3}
21	Alkalinity	[Alkalinity]	mol L^{-1}

2.2. *The transport model framework and its linking with hydrodynamics*

ESTAS-AQUABC is a general box modelling software, which uses the water flow time series amongst the model boxes to calculate the advection as a transport process. The flow time series could be obtained from any source and is not hydrodynamic model dependant. In some simple problems such as analysis of a small and completely mixed reservoir hydrodynamic modelling may not be required at all, or for a river system with limited branching just hydrological routing calculation would produce accurate enough time series for advection. However, more complicated ecological modelling problems require multi-dimensional hydrodynamic modelling to produce the necessary flow time series through box cross sections.

Multi-dimensional hydrodynamic modelling codes provide comprehensive outputs at spatially and temporally high resolution, where each code has its specific output format. To adapt these outputs to a spatially lower resolution model with its own time series format manually would be unproductive work. Therefore, a SHYFEM-specific tool that does this work automatically, was developed. Since the boxes of ESTAS can be of any shape, a finite element hydrodynamic model that could reproduce the outer borders of any model box from hydrodynamic elements would be more suitable than finite difference hydrodynamic code that only supports structured rectangular grids. The main reason for SHYFEM selection was its successful application to many coastal lagoons (also the Curonian Lagoon) and the familiarity of the internals of SHYFEM that were adapted to produce flow time series for the box model.

The hydrodynamic model SHYFEM is a finite element model developed at the CNR-ISMAR of Venice and successfully applied to many coastal environments (Umgiesser et al., 2004; Ferrarin and Umgiesser, 2005; Ferrarin et al., 2010; Bellafiore et al., 2011; De Pascalis et al., 2012; Zemlys et al., 2013). The model is freely available on the SHYFEM web page: http://www.ismar.cnr.it/shyfem. Detailed description of model hydrodynamic and transport equations and its application to the Curonian Lagoon is given in Zemlys et al. (2013) and Umgiesser et al. (2016). Finally, a MATLAB based tool was developed to produce many of the ESTAS-AQUABC required input files as illustrated in Fig. 1.

2.3. Pelagic ecological model

The pelagic ecological model called AQUABC (AQUAtic Biogeochemical Cycles) is a fully featured eutrophication model comprising nutrient cycles, primary production, organic matter mineralization and a zooplankton sub-model tracking the variable stoichiometry. The processes related to state variables in AQUABC are illustrated in Fig. 2. The

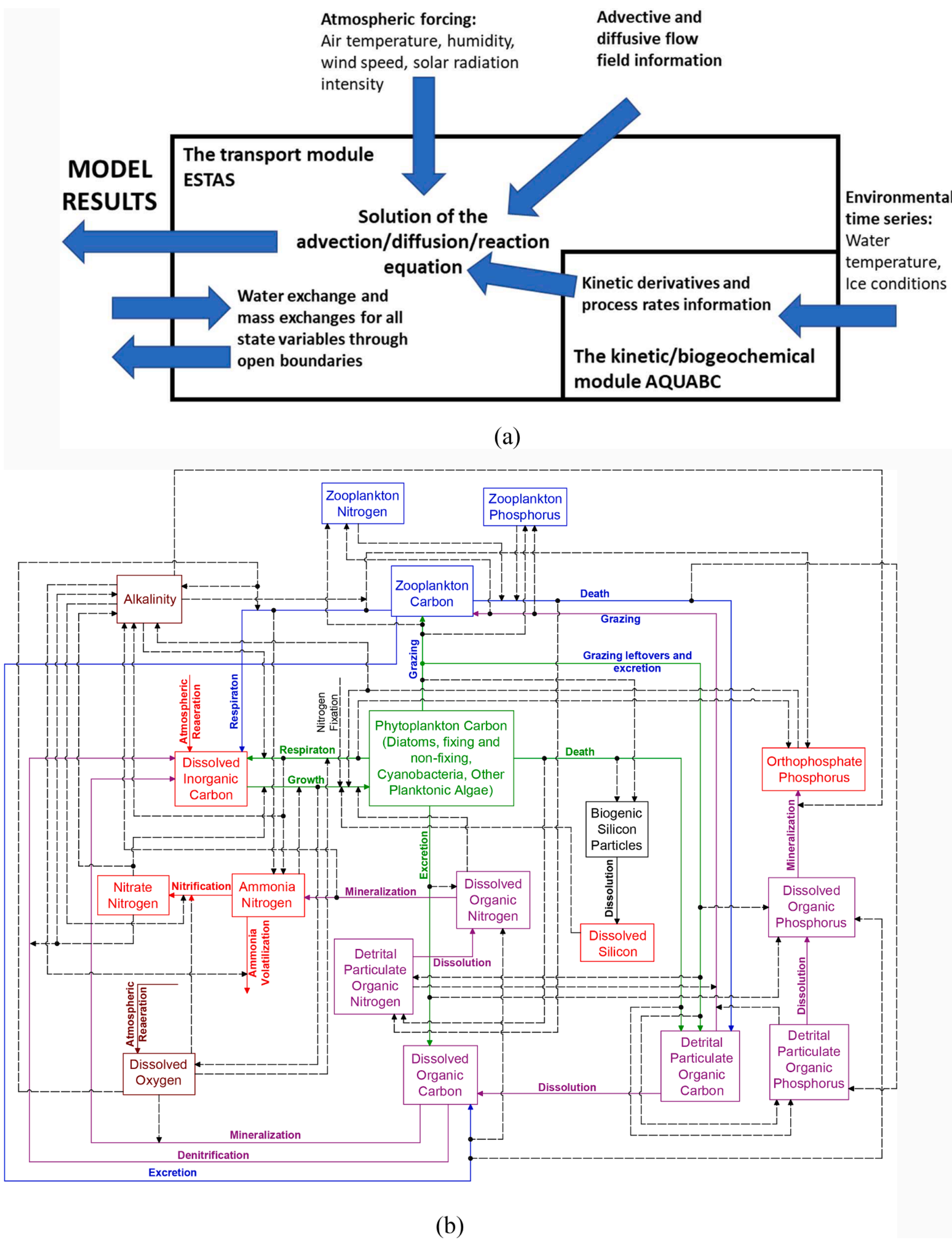

Fig. 2. AQUABC (a) Location within the transport framework (b) State variables and their interactions.

equations describing the kinetics for each state variable are given in Supplementary Material A with the individual equations for each process rate and other details.

The model receives open boundary information, environmental time series, and meteorological time series as forcing inputs. The environmental time series consist of water temperature, which significantly influences various process rates, as well as the saturation concentrations of NH_4—N, carbon dioxide (CO_2), and O_2 in water (see Supplementary Material A for more details). Additionally, the model considers ice conditions, which impact gas transfer between the waterbody and the atmosphere, thereby affecting dissolved O_2, CO_2, and ammonia (NH_3) by partially or completely blocking the water surface from the atmosphere (Supplementary Material A). The meteorological time series utilized for the pelagic ecology model include air temperature, humidity, solar radiation, and wind speed. These meteorological inputs are typically provided to the model as hourly time series. Solar radiation information is particularly utilized for processes related to primary production, while other meteorological variables influence the gas transfer between the water and the atmosphere. Further details on these relationships can be found in Section A.8 (Equations A.8.2 - A.8.11) for dissolved O_2 and the general aeration coefficient, Section A.6.1 (Equations A.6.1.6 - A.6.1.7) for NH_3, and Section A.6.5 (Equations A.6.5.2 - A.6.3) for CO_2. These equations are provided within the electronic Supplement A.

To calculate pH from dissolved inorganic carbon, alkalinity, water temperature, phosphate phosphorus and salinity, CO2SYS model was utilized (Lewis and Wallce, 1998). FORTRAN module of CO2SYS for ESTAS-AQUABC was obtained by converting from MATLAB to FORTRAN an open-source version by van Heuven et al. (2011).

3. Case study application

3.1. Study site

To demonstrate the capacity of ESTAS-AQUABC model we applied it to a case study in the Curonian Lagoon. This lagoon is a large (1584 km^2), shallow (mean depth 3.8 m), and mostly freshwater estuarine system connected to the south-eastern part of the Baltic Sea by a narrow strait (Fig. 3).

The dynamics is dominated by the river discharge with a climatological average of 21.8 km^3/yr (700 m^3/s) (Jakimavicius, 2012). More than 90% of this amount is contributed by the Nemunas River that

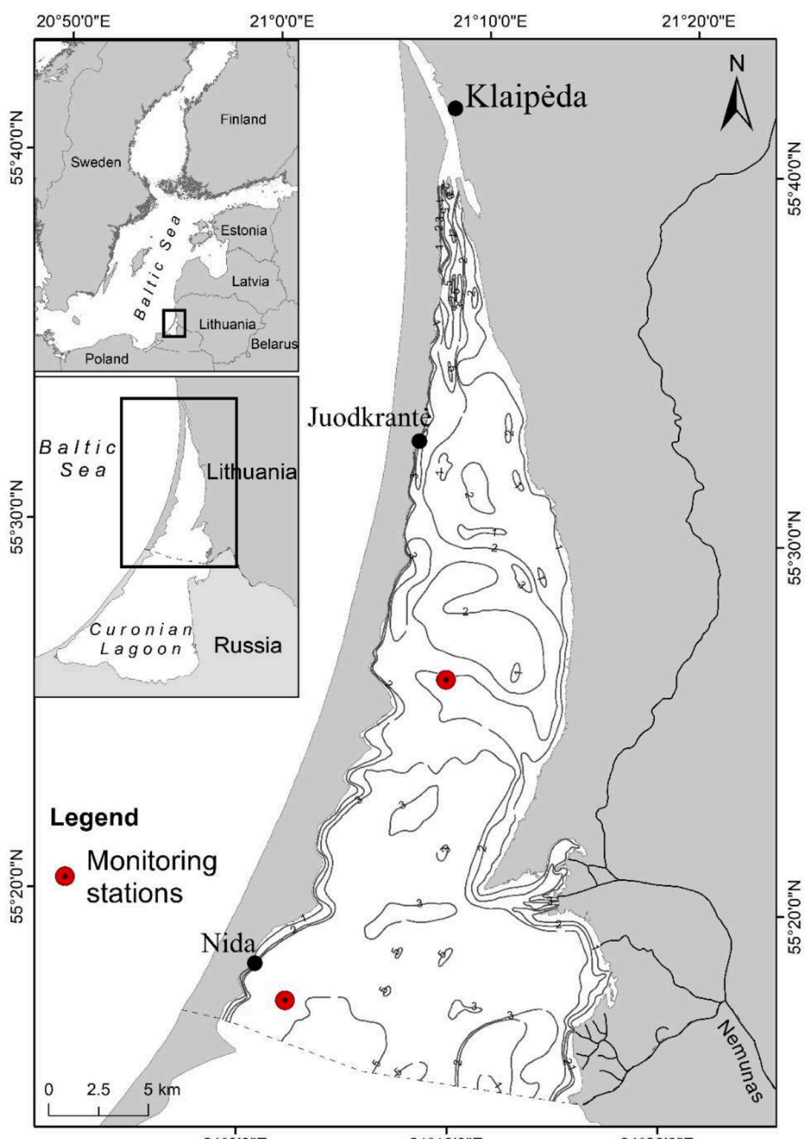

Fig. 3. Map of the Curonian Lagoon (Lithuanian part) showing depth contours. The red circles indicate the water monitoring stations.

discharges into the central and northern part of the lagoon, and all rivers account for 96% of the total freshwater inputs (Jakimavičius and Kraučiūnienė, 2013). Every year the rivers carry the amount of fresh water about 4 times the lagoon volume. Therefore, the southern and central parts of the lagoon are fresh water (average annual water salinity is 0.08 ‰), while the northern part has an average annual water salinity of 2.45 ‰, with irregular salinity fluctuations of up to 7‰ due to Baltic water intrusion (Dailidiene and Davuliene, 2008).

The lagoon occasionally receives inputs from the Baltic Sea through the small opening of the Klaipeda Strait during periods of wind-driven tidal forcing (Zemlys et al., 2013). These events are typically of short duration and result in small increases in average salinity (typically by 1–2 PSU) in the northern portion of the lagoon. Runoff to the Curonian Lagoon exhibits a seasonal pattern, peaking with snowmelt during March and April (Vybernaite-Lubiene et al., 2018). The flow of the Nemunas divides the lagoon into northern-like transitional and central-southern-like confined areas that differ in depth and sediment composition (Zilius et al., 2014). Therefore, the water renewal time is spatiotemporal variable varying from 75 days in northern part to 150 days in central-southern part and around 200 days for the southern part (Umgiesser et al., 2016).

Seasonal water temperature dynamics is typical for shallow temperate lagoons and show a maximum of up to 25 °C (Zilius et al., 2018, 2021). The strait is permanently ice-free, whereas the rest of the lagoon has ice cover during winter (Idzelytė et al., 2019). During ice-free period the lagoon is vertically well mixed owing to the shallow depth and weak salinity gradients (Zilius et al., 2014). Dissolved O_2 concentration fluctuates spatially and temporally (both diurnally and seasonally) (Zilius et al., 2014).

The Curonian Lagoon is characterized by eutrophic conditions with a succession of algal blooms transitioning from diatom and green algae dominated in spring, and to cyanobacteria-dominated blooms in summer (Pilkaityte and Razinkovas, 2007; Zilius et al., 2018, 2021). The shift in phytoplankton community composition coincides with seasonal changes in nutrient ratio in the inputs to the lagoon (Pilkaityte and Razinkovas, 2007). In general, a large excess of N in autumn, winter, and spring has been found, while a marked N deficiency (and to a minor extent of Si) exists in summer (Vybernaite-Lubiene et al., 2017). Seasonal changes in the stoichiometry of nutrient inputs and the occurrence of algal blooms affected nutrient cycling within the lagoon (Petkuviene et al., 2016; Vybernaite-Lubiene et al., 2017; Zilius et al., 2018; Bartoli et al., 2021; Vybernaite-Lubiene et al., 2022). Summer cyanobacteria blooms in the Curonian lagoon diminish its capacity to retain N by altering the fate of algal-N (favouring export over sedimentation) and increasing bioavailability of N via fixation (Zilius et al., 2018).

A hydrodynamic model simulating the area of the Curonian Lagoon and adjacent coastal area of the Baltic Sea was spatially discretised into the mesh with triangular elements having in total 2027 nodes (Fig. 4a). This mesh was used to create the ESTAS model boxes consisting of 25 boxes, which were numbered by SHYFEM mesh utility (Fig. 4b). Model open boundaries were set into the Baltic Sea and the Nemunas River. Boundary concentrations for the model were provided by

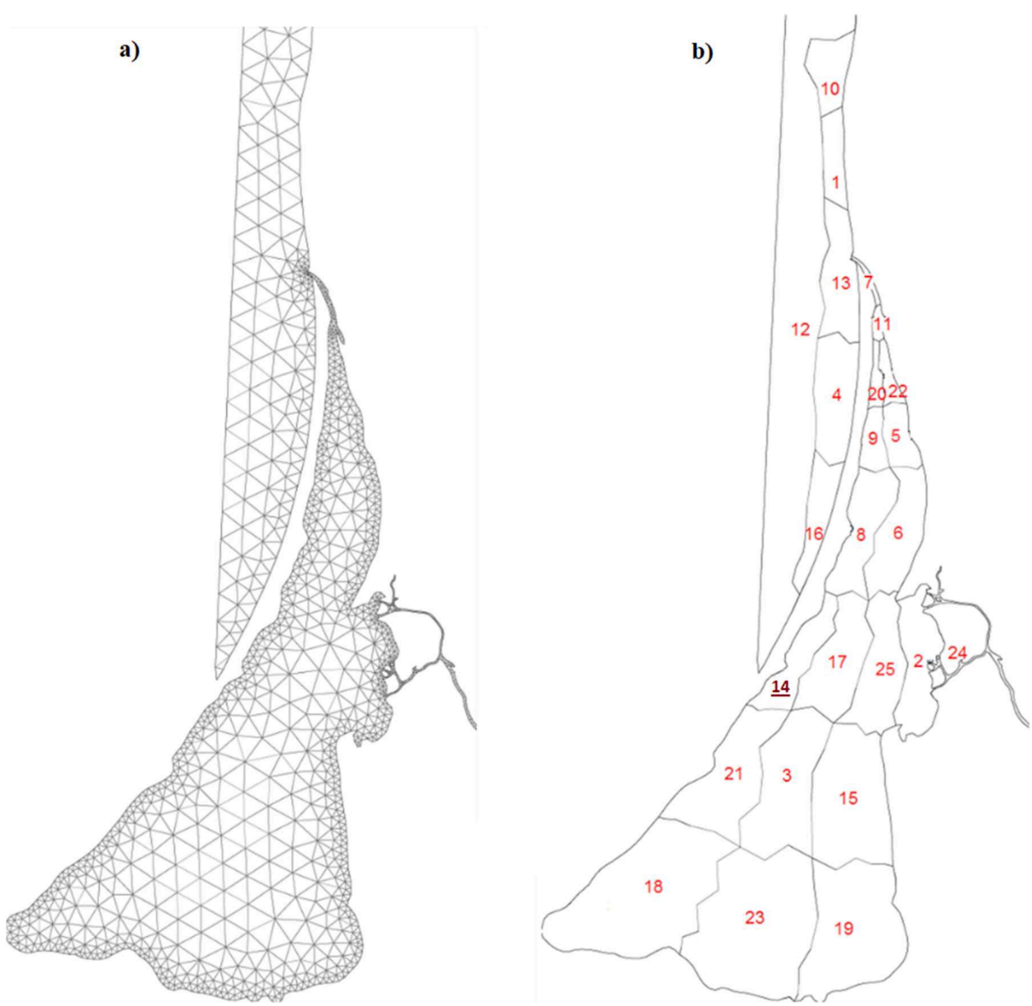

Fig. 4. Spatial discretization of the box-model based on the 2-dimensional finite element hydrodynamic SHYFEM model set up with mesh (a) used in the case study and numbered boxes for ESTAS model (b).

Environmental Research department of Environmental Protection Agency of Ministry of Environment. Since the case study is focused on a shallow lagoon, we used SHYFEM with a 2D setup for the first test application of ESTAS-AQUABC.

The model domain consists of three regions; the Baltic Sea, the northern transitional part and the central-southern stagnant part of the Curonian Lagoon. Even though the Baltic Sea part is not the focus of our study, the model still needs proper boundary conditions for ecological variables which cannot be properly obtained at the Baltic Sea entrance (Klaipėda strait). The conditions are poorly predicted due to the formation of plumes, making it challenging to acquire a high-resolution time series of boundary concentrations. Characterizing this process is particularly complex due to the Klaipėda Strait being an active waterway. This is the reason that the open boundaries of the model were constructed outside the lagoon, in the Baltic Sea where there are calmer conditions in terms of ecological state variables and a monthly time series obtainable from the Environmental Protection Agency would be considered to be adequate for the first application of the model. Considering Fig. 4, Box 12 serves as a general outer boundary condition, whereas boxes 1, 4, 10, 13 and 16 provide a sponging of the boundary condition into the real area of interest. The Curonian Lagoon itself consists of two zones, a northern transitional zone mostly covering the Lithuanian territory of the lagoon and the southern stagnant zone mostly covered by the territory of the Russian Federation. Ferrarin et al. (2008) provides the ideas of hydraulic zoning of the Curonian Lagoon whereas Vybernaite-Lubiene et al. (2021) discusses its biogeochemical dynamics. The transient part consists of box 7 and 11 that represent the Klaipėda strait along with the deeper part covering the harbour area, boxes 20, 22, 9, 5, 8 and 6 covering the relatively wide and shallow part of the transient zone with the Nemunas main branch. Furthermore, boxes 24, 2, 25 and 17 are representing the Nemunas dominated zone, interfacing the transitional zone with the main river flow. One of the monitoring

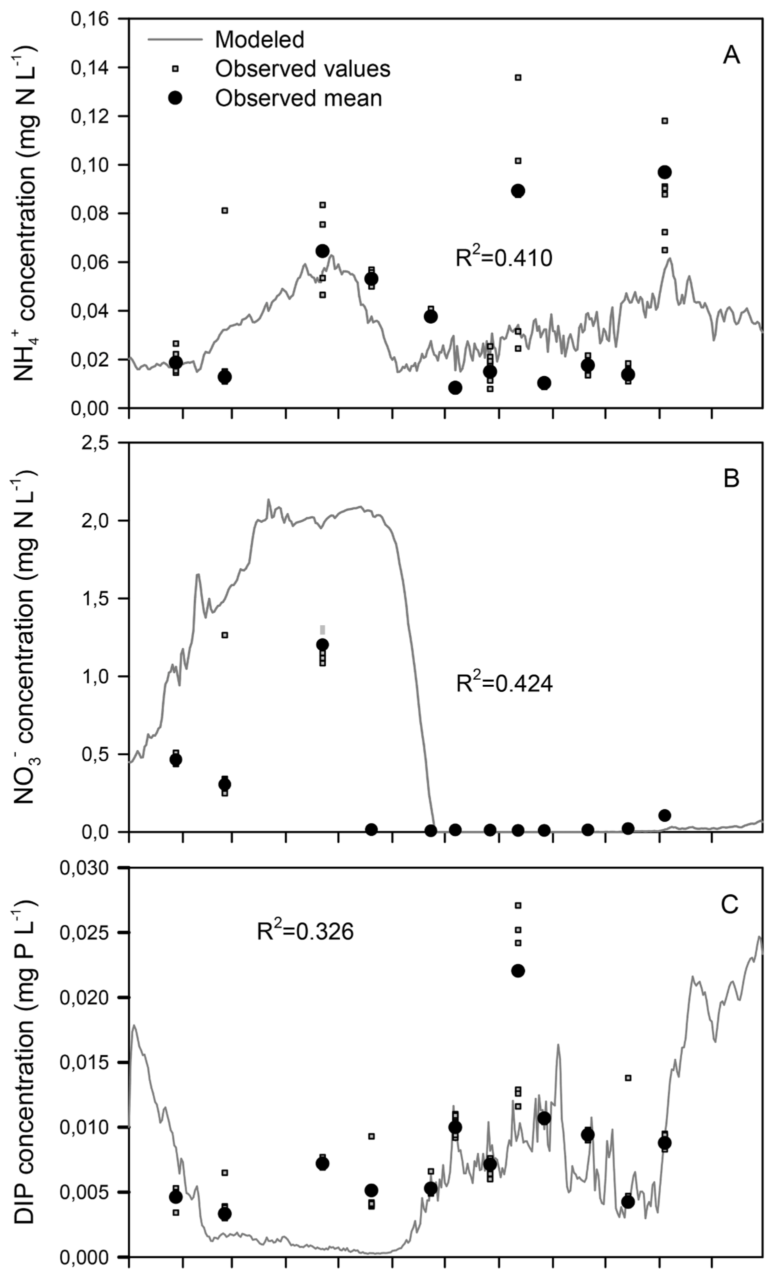

Fig. 5. Modelled (line) and observed (closed circle and square) concentrations of dissolved ammonia (A), nitrate (B) and inorganic phosphorus (C) at the monitoring site in the Curonian Lagoon during 2015. Mean values are based on 6 replicates.

stations is located in Box 8. Remaining boxes represent the deep and stagnant zone mostly in Russian Federation territory. It is important to note that for the Lithuanian territory obviously more data is available. For the southern stagnant part almost no data was available except some indirect data published as maps or averaged values in some papers (Alexandrov, 2010; Alexandrov et al., 2018).

3.2. Model calibration

Since ESTAS-AQUABC allows for a user-entered set of kinetic and stoichiometric model constants, model calibration is possible. A set of simulations, each with a time step of 15 min were conducted for model calibration. Model calibration utilized the available observation data measured at 2 stations (one located in northern transitional (box no. 8), and in central-southern (box no. 14) in January–November 2015 (Zilius et al., 2018). As the ESTAS-AQUABC model is a box model, the calibration procedure comprised two boxes (Fig. 4), where the water column was monitored. In this case study, the model was manually calibrated using the coefficient of determination (R^2) calculated between measured and modelled values as an objective optimization function. To start with manual calibration we have performed single parameter sensitivity runs to figure out what parameters are the most significant for state variable change, also getting approximate values for the calibration steps. In our calibration process, we focused on state variables, related to the nutrient concentrations (NH_4^+, NO_3^-, and PO_4^{3-}) and total phytoplankton carbon as these are expected to be important in such system as the Curonian Lagoon. The calibrated model constants are given in Supplementary Material B (altogether 183 constants). To simplify model calibration R (R Core Team, 2019) based graphical user interface application was developed. The application allows to choose a model constant to calibrate, and produces simulations with different constant values set by the modeller. After the simulations are done, the application produces graphs of state variables with both modelled and measured data.

The calibrated model time series graphs of eutrophication process parameters (NH_4–N, NO_3–N, PO_4-P, N_2 fixation, total phytoplankton biomass) for the model box 14 for 2015 are presented in Fig. 5–7, together with liner relationship between ratio of atmospheric nitrogen (N_2) fixation and heterocystous cyanobacteria biomass, and total dissolved nitrogen in Fig. 8.

3.3. Spatial analysis

In the ESTAS-AQUABC box model, where each box is represented as a completely mixed reactor, no box-internal spatial concentration gradients exist, and the typical interpolation methods such as kriging are not appropriate. For the spatial analysis, based on the modelled data of each 25 boxes, values were arranged by seasons: winter (January–February), spring (March–May), summer (June–August), and autumn (September–November). For the spatial representation of N_2 fixation and net primary production in the Curonian lagoon (19 boxes out of 25 boxes in the model domain) the sum of seasonal values of each box were used, and a classification method using QGIS 3.4 program was applied (Fig. 9 and 10).

4. Results and discussion

The model was applied to the Curonian Lagoon on a fairly complex model domain consisting of 25 model boxes (Fig. 4). To address the computational burden associated with a spatially high-resolution model, we have conceptualized the model as a box model. Initial tests have indicated that even when running on a server, a one-year 2D simulation on a relatively simple mesh with 1300 nodes and 2000 elements generated for the Curonian Lagoon takes several hours, considering hydrodynamic simulation, salinity, and temperature, while excluding water quality state variables. In a similar study conducted by Zemlys et al. (2008), which employed a simpler water quality model with only 9 state variables compared to the 21 state variables in our study, several hours were also required for a single set of simulations for one year. Given that the model calibration process necessitates hundreds of simulations, it is more practical to utilize a faster-performing model. The box modelling tool developed in this study enables a one-year simulation of the same 25-box model domain in approximately two minutes, even when executed on a single processor (4th Gen i7). Moving forward, our future development will focus on finalizing the incorporation of AQUABC into SHYFEM and developing the necessary tools and utilities to facilitate the transfer of calibrated model parameters into the SHYFEM framework.

AQUABC has several features that are rarely considered in other modelling studies (see Table 2). For example, N_2 fixation that was only considered by Ahlvik et al. (2014), Robson et al. (2008), Hense and Beckmann (2010) and Zhu et al. (2016). Inorganic carbon cycle was only considered by Hull et al. (2008), Rodrigues et al. (2009) and Zouiten et al. (2013). Whereas some of the recent models consider alkalinity as a state variable. AQUABC calculates pH, where DIC, alkalinity, water temperature, PO_4-P, and salinity are used. AQUABC has also another unique feature, not considered by any other models listed in Table 2, which is the hypoxia stress on organisms. Another process that is not found in most of the eutrophication models is NH_3 outgassing from water column to the atmosphere, called ammonia stripping, which could be potentially important pathway in eutrophic estuarine systems where due to active photosynthesis high pH prevails. More detailed

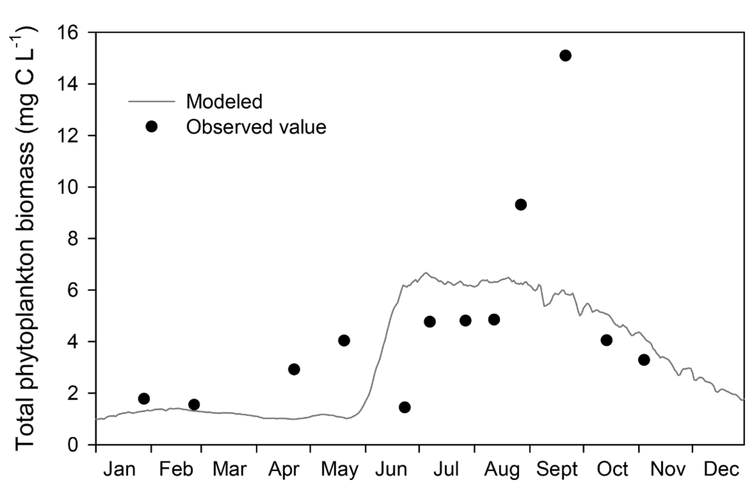

Fig. 6. Modelled (line) and observed (closed circle and square) total phytoplankton biomass at the monitoring site in the Curonian Lagoon during 2015.

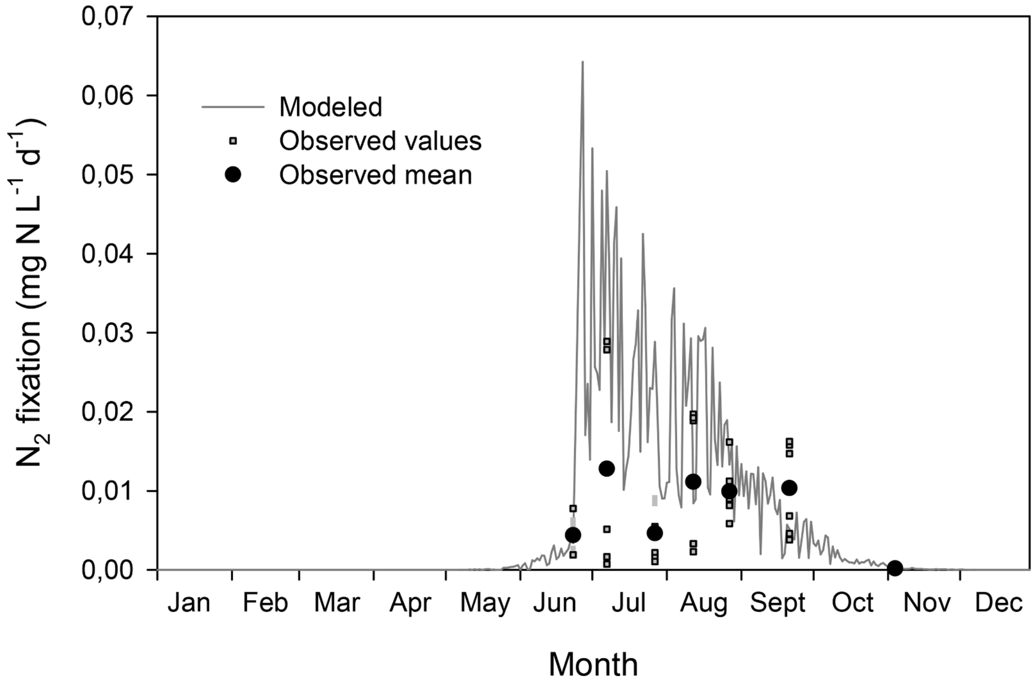

Fig. 7. Modelled (line) and measured (closed circle and square) atmospheric nitrogen at the monitoring site in the Curonian Lagoon in 2015.

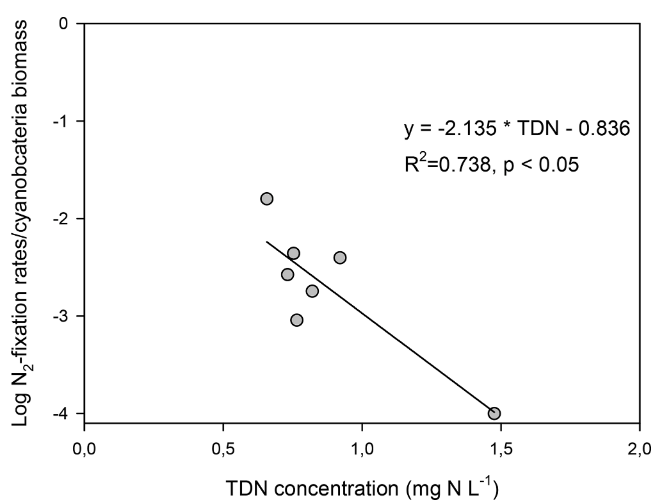

Fig. 8. Liner relationship between ratio of atmospheric nitrogen fixation and heterocystous (nitrogen fixing) cyanobacteria biomass, and total dissolved nitrogen (TDN$=NH_4^+$-N + NO_3^--N + NO_2^--N + DON).

information on the model are given in Supplementary Material A.

The modelling framework developed in present study for the Curonian Lagoon attempts to capture spatiotemporal patterns of pelagic biogeochemical and biological processes driven by seasonally variable freshwater inputs from a large river, brackish water intrusions and metabolic activity (rates). The Curonian Lagoon is characterized by a strong spatiotemporal heterogeneity in both chemical and physical parameters as well as bottom sediment characteristics (Ferrarin et al., 2008; Petkuviene et al., 2016; Umgiesser et al., 2016; Zilius et al., 2018, 2021) affecting in turn seasonal and spatial patterns of biological and biogeochemical processes.

The hydraulic transport calculated with SHYFEM allowed us to follow closely the seasonal circulation patterns in the lagoon including discharges from the Nemunas River and exchange with the Baltic Sea, where the boundary conditions were set up. As the Curonian Lagoon is quite shallow and mostly vertically well-mixed (Umgiesser et al., 2016), the absence of vertical resolution in the modelled boxes is an acceptable simplification of transport processes existing in the lagoon. Thus, each simulation for a one-year period during ESTAS-AQUABC calibration took 75 s on a standard personal computer using the configuration shown in Fig. 2. The calibrated model constants are given in the Supplementary Material B.

The calibration results demonstrate moderate success in following the dynamics of dissolved inorganic nutrients while capturing the main pattern in total phytoplankton biomass (Fig. 5, 6). The coefficient of determination (R^2) values for these variables are well within the acceptable performance for biogeochemical models except for nitrates according to Arhonsitsis and Brett (2004). However, the decline of NO_3—N concentrations in the summer, so typical for the lagoon's biogeochemical cycles, is predicted quite well (Fig.5). In general, the model predictive power is better than in the previous modelling study in the Curonian Lagoon (Zemlys et al., 2008), where a simpler biogeochemical model was applied. Zemlys et al. (2008) calibrated the SHYFEM/EUTRO model (Umgiesser et al., 2003) for the Curonian Lagoon and achieved a coefficient of determination of 0.56 for NO_3—N, which is higher than the value of 0.42 obtained in our study. However, the R^2 reported by Zemlys et al. (2008) for NH_4^+-N and PO_4^{3-}-P were 0.05 and 0.07, respectively, which were significantly lower than the R^2 achieved in present study (0.41 for NH_4—N and 0.33 for PO_4^{3-}-P).

Since the Curonian Lagoon is a shallow system, benthic biogeochemical processes and flux at the sediment-water interface are important regulating nutrient dynamic in overlaying water column, especially in summer when riverine nutrient inputs decrease and water residence time is longer (Petkuviene et al., 2016; Zilius et al., 2018). The benthic fluxes of NH_4^+-N, NO_3^--N and PO_4^{3-}-P measured in the same period (Zilius et al., 2018) could account for at least 50% of NH_4^+-N concentration mismatch between modelled and observed data. Sediments were always a NH_4^+-N source, particularly during the summer when intensive mineralization occurred (Zilius et al., 2018; Bartoli et al., 2021). Although sediments in the lagoon could potentially serve as a sink for NO_3^--N, the measured fluxes at the sediment-water interface via intact core incubations (Zilius et al., 2018), accounts for only a marginal 2% of the discrepancy observed between the modelled and observed concentrations in spring. The rest is probably could be attributed to the pelagic

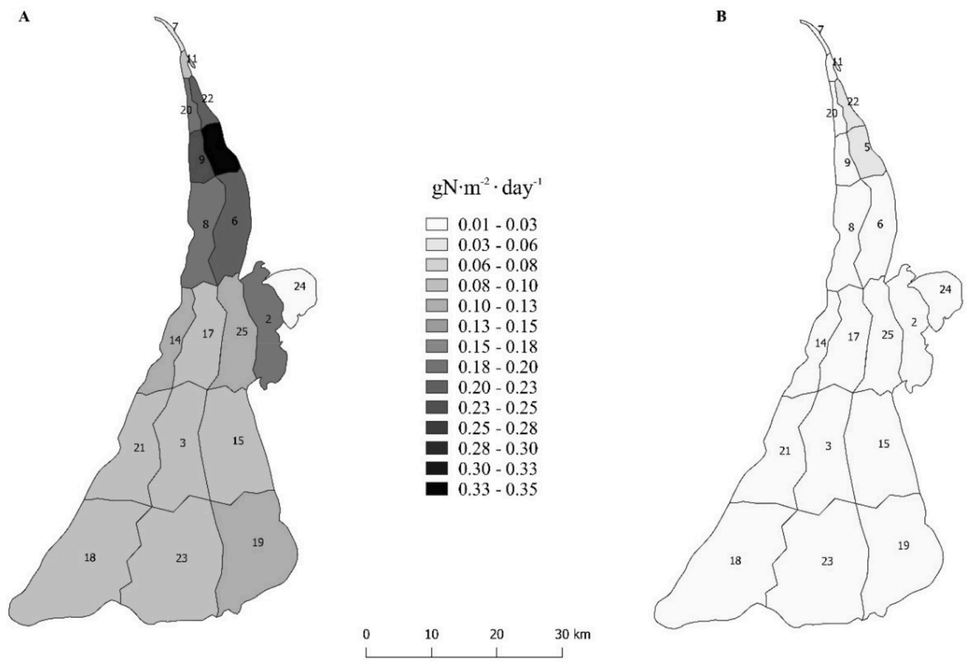

Fig. 9. Spatial distribution of atmospheric nitrogen fixation in the Curonian lagoon in summer (A) and autumn (B) periods in 2015.

uptake which was not measured. Broman et al. (2021) suggest that despite the importance of benthic contribution to water column processes in the lagoon, pelagic recycling of organic matter to ammonia supplies the major amount of required N to maintain high phytoplankton biomass. Thus, N cycling in the ecosystem of the Curonian Lagoon is also crucial especially in summer periods when N limitation persists with low standing concentrations of inorganic nutrients (< 1 μM) and high phytoplankton biomass. Therefore, calibration of related process rates other than N_2 fixation such as assimilative uptake of dissolved organic nitrogen, denitrification and dissimilatory nitrate reduction to ammonium (DNRA) should be considered on further implementations of the pelagic model. Zilius et al. (2018) demonstrated that assimilative uptake by benthic and pelagic diatoms is the most important mechanism regulating NO_3^--N concentrations in overlaying water. However, in absence of a benthic compartment in the model, this flux was left unaccounted as well. The ecological model used in this study has shown some discrepancies in accurately estimating the abundance of phytoplankton and PO_4-P, while it tends to overestimate NO_3^--N levels. Specifically, in August, the model fails to capture the actual higher values of PO_4^{3-}-P, NH_4^+-N, and phytoplankton due to the significant aggregation of cyanobacteria during this period. Unfortunately, the current transport model employed in this study is unable to account for this specific phenomenon. Consequently, the modelled values of phytoplankton biomass underestimate the observed values. The underestimation of PO_4^{3-}-P can be attributed to the data used in the forcing function, which relied on incubation experiments. It should be noted that the redox conditions in these incubation experiments do not perfectly align with the in situ conditions, leading to the observed underestimation of PO_4^{3-}-P values. Thus, the differences in the environmental conditions between the incubation experiments and the natural habitat contribute to the discrepancies in estimating PO_4-P levels in the ecological model. However, it is known that even local hypoxia developed during summer due to short-term thermal stratification (which could be easily missed during measurement campaigns) and intensive respiration could trigger intensive phosphorus release from sediments (Zilius et al., 2014). Moreover, even as we tried to mimic some effects of resuspension by introduction of relationship between bottom shear stress and settling velocities (zero values when critical shear stress is exceeded), the vast majority of resuspension related processes were left unaccounted. The absence of a benthic compartment in the model is a crucial aspect that necessitates careful consideration. It is imperative to evaluate the determination that future iterations of this model should incorporate a comprehensive description of bentho-pelagic relations. In this particular context, it becomes evident that the model tends to overestimate the levels of N precisely when the fauna populations experience growth and proliferation. Simultaneously, the population increase observed during the spring and summer seasons undeniably impacts the zooplankton population, which feeds on phytoplankton, thus explaining the model's underestimation. Furthermore, a noteworthy phenomenon is the migration of a portion of this fauna to the sea, carrying with them the accumulated nutrients (Fernández-Alías et al., 2022).

One of the specific objectives of this work was to follow the nitrogen cycle-related processes in a shallow coastal lagoon and, specifically, estimate biological N_2 fixation from atmosphere. It has already been shown by Asmala et al. (2017) and Zilius et al. (2018) that coastal lagoons could retain N on their pathway to the Baltic Sea. Dissolved N can be attenuated by incorporation of dissolved forms into biomass of primary producers (microphytobenthos and phytoplankton) and their settling (burial in sediments) or loss to the atmosphere via denitrification or anammox (Nedwell et al., 1999; Sundbäck et al., 2004; Brion et al., 2008; Zilius et al., 2018; Broman et al., 2021). According to previous studies (Zilius et al., 2018, 2021), favourable conditions are provided for a diazotrophic community during summer and fall with the decrease in NO_3^--N inputs after spring snowmelt in river discharge. However, nitrogen retention can be frequently offset by biological N_2 fixation during summer blooms (Vybernaite-Lubiene et al., 2017). Nevertheless, the importance of this process as an external N source, that may shift the role of the lagoon as a sink to source, remains vaguely quantified or numerically assessed. Measured rates of biological N_2 fixation show that it is relevant and represents significant input of N to the southern half of the lagoon during the summer periods (Zilius et al., 2018, 2021). Even though, N_2-fixation rates based on direct measurements are sparse, highly variable, and restricted in spatial resolution, lagoon-scale estimates of N_2 fixation derived by remote-sensing-based estimates of chlorophyll a show spatially heterogeneity especially in fall (Zilius et al., 2021). Our developed model allowed us to assess the N_2 fixation patterns along the spatial and temporal scales and contribute to

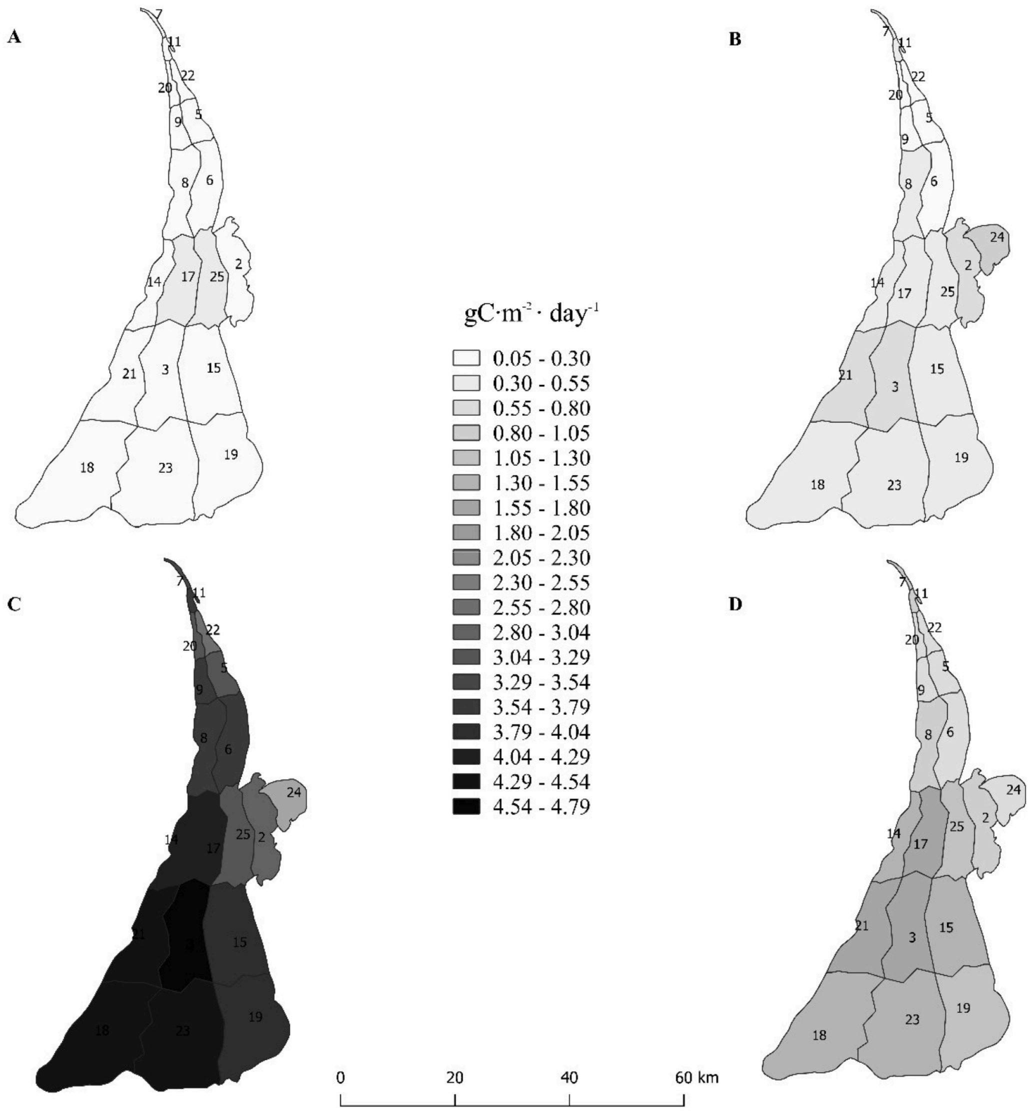

Fig. 10. Spatial distribution of net primary production in the Curonian lagoon during winter (A), spring (B) summer (C) and autumn (D) periods in 2015.

the understanding of factors regulating this process in estuarine environments. The modelled results agree qualitatively with observations (R^2=0.3 without removing the outliers in the data and 0.36 after the outliers are removed $p < 0.05$). Moreover, we tested the model assumptions linking N_2 fixation to the relative abundance of total dissolved nitrogen (TDN = NH_4^+-N + NO_3^--N + DON). It shows that TDN depletion drives both atmospheric N_2 assimilation and total abundance of N_2 fixing cyanobacteria (proved by the analysis of in situ measured data (R^2= 0.738, $p < 0.05$)). According to field measurements the N_2 fixation rate accelerates steeply at TDN concentrations below < 0.8 mg N L^{-1} following the linear relationship (Fig. 8). The dynamic of total dissolved N is primarily associated with river discharge, depending on the combination of factors such as precipitation, temperature variability, and vegetation activity in the river catchment (Vybernaite-Lubiene et al., 2018), which, in turn, are included in the modelling framework as external inputs.

Although models are frequently constrained by uncertainties and/or requires further improvements, they could provide invaluable tool for nutrient mass balance calculations over temporal and spatial patterns. Momentary measurements always represent temporal and spatial snapshots of processes (e.g. N_2 fixation), while their dynamics are often in scale of hours and spatially extremely heterogenic. We attempted to upscale N_2 fixation and net primary production processes to the spatial scale of the lagoon. Mean N_2-fixation rates varied by one order of magnitude amongst boxes with mean rates of 0.026±0.008 g N $m^{-2}d^{-1}$ and 0.002±0.001 $m^{-2}d^{-1}$ in summer and autumn, respectively. Comparison of these modelled results for 18 boxes inside the lagoon with averaged measured values (see to Zilius et al., 2018, 2021) shows that model rates are 42% higher in summer while 22% lower in autumn. These differences can be attributed to the spatial heterogeneity caused by patchy distribution of N_2 fixing cyanobacteria over lagoon surface, resulting in different diazotrophic activity (Zilius et al., 2021). Satellite images revealed quite heterogeneous distribution of phytoplankton concentrations indicating strong effect of meteorological conditions (Bresciani et al., 2014; Vaičiūtė et al., 2021). Our previous study demonstrates that remote sensing-based estimates of N_2 fixation for the Curonian Lagoon (mean = 2.1 ± 0.1 mmol N $m^{-2}d^{-1}$) align closely with experimental measurements obtained at the two stations, which were subsequently upscaled to represent the entire lagoon (mean = 1.9 ± 0.9 mmol N $m^{-2}d^{-1}$; Zilius et al., 2021). Also, at this stage of model development, these patches cannot be reproduced, since the model developed in this study is a depth-integrated model which considers the water column as vertically mixed whereas Bresciani et al. (2014) and Vaičiūtė et al. (2021) results are related to the water surface and to scum-like formations in some cases. To produce such spatial patches, a vertically layered version of the model's input data sets would be required, which

Table 2
Models used in Eutrophication Studies.

Reference	Spatially high-resolution models in two or more dimensions	Number of phytoplankton compartments	Number of cyanobacteria compartments	Nitrogen fixation included	Macroalgae or submerged aquatic vegetation included	Coupling with sediments	Zoo-plankton included	Fish or other macro animals included
Hieronymus et al., 2021	Yes	0	1	Yes	No	No	No	No
Vichi et al., 2020	Yes	4	0	No	No	Yes	Yes	No
Baustian et al., 2018[1]	Yes	6	2	Yes	Yes	No	No	No
Aumont et al., 2015	Yes	2	0	Yes	No	No	Yes	No
Dugdale et al., 2016	No	1	0	No	No	Yes	Yes	Yes
Zhu et al., 2016	Yes	2	1	Yes	No	Yes	Yes	No
Cosme et al., 2015	Yes	1	0	No	No	No	Yes	Yes
Ahlvik et al., 2014	No	1	1	Yes	No	No	No	No
Cossarini et al., 2014	Yes	0	0	No	No	No	No	No
Turner et al., 2014	No	1	0	No	No	No	Yes	No
Zouiten et al., 2013	Yes	1	0	No	No	No	Yes	No
Fernandez et al., 2012[2]	Yes	1	0	No	No	Yes	Yes	No
Ruzicka et al., 2011	No	2	0	No	No	No	Yes	No
Cerco et al., 2010b	Yes	2	1	No	Yes	No	Yes	Yes
Chao et al., 2010[3]	Yes	1	0	No	No	No	No	No
Estrada and Diaz, 2010[3]	No	2	1	No	No	No	No	No
Fennel, 2010	No	1	0	No	No	No	Yes	Yes
Hense and Beckmann, 2010[4]	No	0	2	No	No	No	No	No
Hochard et al., 2010	No	1	0	No	Yes	Yes	Yes	No
Jia et al., 2010	No	3	1	No	No	No	Yes	No
Llebot et al., 2010[3]	No	2	0	No	No	No	Yes	No
Lopes et al., 2010	Yes	1	0	No	Yes	No	No	No
Serizawa et al., 2010	No	0	1	No	No	No	No	No
Timmermann et al., 2010	Yes	1	0	No	No	Yes	Yes	No
Zhang and Li, 2010	No	1	0	No	No	Yes	No	No
Chung et al., 2009[5]	No	1	0	No	No	Yes	No	No
Rodrigues et al., 2009	Yes	1	0	No	No	No	No	No
Sheng and Kim, 2009	Yes	2	1	No	No	Yes	Yes	No
Hull et al., 2008	No	1	0	No	No	No	No	No
Robson et al., 2008	Yes	4	1	Yes	No	No	No	No
Sohma et al., 2008[5]	No	1	0	No	No	Yes	Yes	No
Trolle et al., 2008[5]	No	2	0	No	No	No	Yes	No
Ménesguen et al., 2007	Yes	2	0	No	No	Yes	Yes	No
Vichi et al., 2007	No	4	0	No	No	No	Yes	No
Laurent et al., 2006	No	1	0	No	No	No	No	No
Nobre et al., 2005[6]	No	1	0	No	Yes	No	No	Yes
Zhou et al., 2005[7]	No	0	0	No	No	Yes	No	No
Chau, 2004	Yes	1	0	No	No	No	No	No
Hartnett and Nash, 2004	Yes	1	0	No	No	No	No	No
Hongping and Young, 2003	No	1	0	No	No	Yes	Yes	Yes

(continued on next page)

Table 2 (*continued*)

Reference	Spatially high-resolution models in two or more dimensions	Number of phytoplankton compartments	Number of cyanobacteria compartments	Nitrogen fixation included	Macroalgae or submerged aquatic vegetation included	Coupling with sediments	Zoo-plankton included	Fish or other macro animals included
Wang et al., 2003[7]	No	0	0	No	No	Yes	No	No
Umgiesser et al., 2003	Yes	1	0	No	No	No	Yes	No
Melaku-Canu et al., 2003	Yes	1	0	No	No	No	Yes	No
Melaku-Canu et al., 2001	Yes	0	0	No	No	No	No	No

[1] Refer to Los, 2009 for further details.
[2] The eutrophication model is developed for a single compartment of phytoplankton, however the phytoplankton community in the study area is strongly dominated by filamentous cyanobacteria (over 80%) according to Miracle et al. (1984).
[3] The effects of sediment on the water column are considered with fluxes, however sediments and water column were not coupled with a closed mass balance.
[4] Includes a growth cycle model of a single cyanobacteria group with multiple compartments representing the life cycle.
[5] The model used in the study is more comprehensive, however it was utilized with a limited number of options.
[6] The model domain has a simple representation; however, it is linked with a high-resolution hydrodynamic transport model.
[7] No water column, sediment only.

were not available at the time of this study. However, it is important to note that our model is fully capable of incorporating such simulations, and we recognize the potential for future studies to address this gap. As more comprehensive data, including the southern part of the lagoon within the Russian Federation territory, becomes available, it would be valuable to further enhance the model infrastructure by incorporating an option to track scum patches formed on the water surface, considering the influence of wind drift.

In general, our results from upscaling show that mean daily contribution of biological N_2 fixation 50% exceeded one provided by summer riverine inputs (0.013 g N $m^{-2} d^{-1}$), which is line to our recent findings (e.g., Zilius et al., 2021). This indicates the importance of atmospheric N_2 fixation, particularly during summer when the riverine discharges are lowest. In addition, we estimated (R^2=0.15 without removing the outliers in the data and 0.26 after the outliers are removed $p < 0.05$) the net primary production which is the difference of gross primary production and respiration. The spatial patterns of net primary production distribution (Fig. 10) provided better understanding heterogeneity of phytoplankton-based processes in the Curonian Lagoon. Obtained results indicates highest production rates observed in summer (0.96±0.45 g C $m^{-2} d^{-1}$) and autumn (0.13±0.07 g C $m^{-2} d^{-1}$). The primary production (PP) in Nida station is estimated to be approximately 13 mmol $m^{-3} h^{-1}$, consistent with earlier measurements that indicated the highest PP rates in the southern region of the lagoon (Zilius et al., 2014). Obtained results for net primary production allow estimation of biological N assimilation by using measured molar carbon and N ratio. Our calculated mean assimilative N uptake values vary from <0.01 to 0.19 g C $m^{-2} d^{-1}$ with peak in summer. There is clear decoupling between assimilative N uptake and supply in time, the latter being the highest in spring. So, the model allows us to trace the further fate of assimilated nitrogen – whether it is remineralized to support autotrophic phytoplankton development.

The differential behaviour exhibited by the model concerning nitrogen and phosphorus can be attributed to the varying ratios between these elements within the relevant compartments, exerting a more significant influence on one element over the other. This ratio, or the proliferation of microbial organisms with dissimilar elemental content ratios in comparison to phytoplankton, may serve as a fundamental factor contributing to anomalies and hypoxic events, as elucidated in the study conducted by Fernández-Alías et al. (2022). Moreover, it is important to note that the occurrence of a "clearwater" phase in the Curonian Lagoon, where top-down control assumes relevance in regulating the system dynamics, is limited to a relatively brief period.

5. Conclusion

A new, high performance and open-source modelling software package was developed to simulate the biogeochemical cycles of nutrients and the eutrophication process for estuaries and coastal lagoons. The software package consists of three main components:

- R-based GUI application that acts as a workbench helping the users to calibrate the model and view the model results with some simple post processing.
- MATLAB based scripts that read the information written by SHYFEM and built most of the model inputs automatically.
- The computational engine of the model developed in FORTRAN modularly, with two sub-components: i) a transport component that solves the advection-diffusion-reaction equation discretized as the box model domain, and therefore handles the material fluxes between the boxes, and ii) a kinetic component that provides the transport component with reaction rates in each model box. This design allows the modeller to create new models for different aquatic ecosystems just by modifying the kinetic component or removing it completely and writing a new subroutine to simulate a different type of ecosystem. The model is not conceptualized as a hydrodynamic model. The water fluxes within the model boxes should be calculated/estimated externally by externally such as the application of a hydrodynamic model, extract resources and format them as time series acceptable for ESTAS-AQUABC. Salinity and temperature are handled the same way.

ESTAS-AQUABC is designed in a way to be flexible in both viewpoints: i) model domain configuration and ii) modification/integration of new pelagic kinetic sub-models into the box model approached solver for the transport equation. The software developed in this study is not only suitable to simulate biogeochemical processes in coastal lagoons but it is also appropriate for inland waters (lakes, rivers) though some efforts would be needed to conduct more precise simulations in rivers. All the components of the modelling package are open-source software to ensure accessibility and easy modification. At this stage the model is ready to be provided to other users for free with a research-partnership contract. The model itself is capable to simulate such parameters and processes as alkalinity and atmospheric N_2 fixation, which are rarely simulated in estuarine NPZD models.

A test case study was conducted on the Curonian Lagoon, with an acceptable level of preliminary calibration which provided satisfactory results for the dynamics of inorganic nutrients and, particularly, pelagic N_2 fixation. It could be concluded that most of the discrepancy between modelled and measured data could be attributed to the incapability of present model to simulate benthic compartment and bento-pelagic coupling, including the resuspension, settling and burial of diatoms. This knowledge gives us clear direction where and how to improve the model to provide better prediction power and capacity. Apart from these necessary enhancements there is increasing evidence that not only filamentous cyanobacteria, but also heterotrophic unicellular microorganisms are capable to fix N_2 (e.g., Bentzon-Tilia et al., 2015; Zilius et al., 2020), which must be reflected in the future by adding additional state variable as different size groups of N_2-fixers in the kinetic part of the modelling framework.

Software availability

ESTAS/AQUABC model was developed by Ali Ertürk, Petras Zemlys and Arturas Razinkovas with contributions of Georg Umgiesser. It is a free and opensource program which can be downloaded via the GitHub repository [https://github.com/aquabc/estas_aquabc-model]. ESTAS/AQUABC model was initially released in 2022 and was developed to run on Windows and Linux; the authors have not tested it on Macintosh operating systems. It was written entirely in FORTRAN 2003. ESTAS/AQUABC model's program files are less than 1.47 MB. Some external utilities that run under MATLAB and R environment are under development and will be made available when they are ready for general release.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgement

The manuscript preparation was supported by the Research Council of Lithuania (LMT) grant "The role of atmospheric nitrogen fixation in the largest eutrophicated European lagoon (NitFix)" (Agreement No. S-MIP-17–7). We gratefully thank the Environmental Research department of Environmental Protection Agency of the Ministry of Environment for providing meteorological and Nemunas River discharge data.

Supplementary materials

Supplementary material related to this article includes Appendix A., Appendix B and Appendix C. Appendix A. has more detailed description on pelagic ecology model in the ESTAS-AQUABC framework. Appendix B. includes AQUABC model constants and their values after preliminary calibration. Appendix C. includes information on linking ESTAS with hydrodynamic models with focus to SHYFEM hydrodynamic model.

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ecolmodel.2023.110509.

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